PHISHING EMAIL DETECTION & AWARENESS REPORT

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EXECUTIVE SUMMARY

Phishing remains one of the most prevalent and damaging cyber threats
facing businesses today. According to industry reports, over 90% of
successful cyberattacks begin with a phishing email.

This report presents a detailed analysis of phishing email samples, identifying common attack techniques, red flags, and providing practical awareness guidelines that employees and businesses can use to protect themselves.

A total of 4 email samples were analyzed:

- 3 classified as Phishing
- 1 classified as Safe (used as a comparison baseline)

Key findings include:

- All phishing emails used fake or spoofed sender domains
- All phishing emails contained suspicious external links
- All phishing emails used urgency and fear-based language
- None addressed recipients by their real name

This report is intended to serve as both an analytical document and a practical awareness resource for employees and organisations.

WHAT IS PHISHING?

Phishing is a type of cyber attack where criminals send fraudulent emails pretending to be from trusted organisations — such as banks, delivery companies, or internal IT departments — in order to trick recipients into:

- Clicking malicious links
- Entering login credentials on fake websites
- Downloading malware-infected attachments
- Sharing sensitive personal or financial information

WHY IS IT DANGEROUS?

Phishing attacks are dangerous because they exploit human psychology rather than technical vulnerabilities. Even technically skilled users can fall victim to well-crafted phishing emails.

The consequences for businesses include:

- Financial loss through fraud or ransomware
- Data breaches exposing customer information
- Reputational damage
- Legal and compliance penalties (GDPR, PCI-DSS)

TYPES OF PHISHING

- Email Phishing — mass emails targeting many users at once
- Spear Phishing — targeted attack on a specific individual
- Smishing — phishing via SMS text messages
- Vishing — phishing via phone calls
- Clone Phishing — duplicate of a legitimate email with malicious links

METHODOLOGY

This analysis was conducted using the following approach:

1. SAMPLE COLLECTION
   Phishing email samples were collected from publicly available educational repositories and crafted based on real-world phishing patterns documented by security researchers.

2. HEADER ANALYSIS
   Email headers were examined using:
   • Google Admin Toolbox (toolbox.googleapps.com)
   • MXToolbox Email Header Analyzer (mxtoolbox.com)

3. DOMAIN INSPECTION
   Sender domains and linked URLs were inspected to identify:
   • Domain spoofing
   • Lookalike domains
   • Mismatched sender addresses

4. CONTENT ANALYSIS
   Email body content was reviewed for:
   • Urgency and fear-based language
   • Generic or impersonal greetings
   • Suspicious calls to action
   • Requests for sensitive information

5. CLASSIFICATION
   Each email was classified using a three-tier system:
   • SAFE — legitimate email from a verified sender
   • SUSPICIOUS — some red flags but not conclusive
   • PHISHING — confirmed malicious indicators present

Email Sample 1 — Fake Bank Alert

From: security@bankofamerica-alert.com
To: user@gmail.com
Subject: URGENT: Your Account Has Been Suspended

Dear Valued Customer,

We have detected unusual login activity on your Bank of America account.
Your account has been temporarily suspended for your protection.
To restore access immediately, please verify your identity by clicking
the link below:

http://bankofamerica-secure-login.com/verify

You must complete verification within 24 hours or your account
will be permanently closed.

Regards,
Bank of America Security Team
support@bankofamerica-alert.com

Analysis — Email 1 (Phishing):

Indicator	Detail	Risk
Sender Domain	`bankofamerica-alert.com` — not the real domain (`bankofamerica.com`)	🔴 High
Suspicious Link	`bankofamerica-secure-login.com` — fake domain mimicking real bank	🔴 High
Urgency Language	"24 hours or permanently closed" — pressure tactic	🔴 High
Generic Greeting	"Dear Valued Customer" — no real name used	🟡 Medium
Fear Tactic	"Account suspended" — designed to cause panic	🔴 High

Email Sample 2 - Fake IT Support

From: itsupport@company-helpdesk.net
To: employee@targetcompany.com
Subject: Action Required: Password Expiry Notice

Dear Employee,

Your company password will expire in 2 hours.

Please reset your password immediately using the link below
to avoid losing access to company systems:

http://reset-password-portal.net/login

If you do not reset within 2 hours, your account will be
locked and IT will need to manually restore access.

IT Support Team
Helpdesk | Company IT Department

Analysis — Email 2 (Phishing):

Indicator	Detail	Risk
Sender Domain	`company-helpdesk.net` — generic, not a real corporate	🔴 High
Suspicious Link	`reset-password-portal.net` — unrelated to any real company	🔴 High
Urgency Language	"2 hours" — extremely short deadline to pressure quick action	🔴 High
Generic Greeting	"Dear Employee" — no name, no employee ID	🟡 Medium
Impersonation	Pretends to be internal IT department	🔴 High

Email Sample 3 - Fake Package Delivery

From: delivery@fedex-tracking-notification.com
To: customer@gmail.com
Subject: Your Package Could Not Be Delivered — Action Required

Dear Customer,

We attempted to deliver your package today but were unable
to complete the delivery due to an incorrect address.

To reschedule your delivery, please confirm your address
and pay a small redelivery fee of $1.99:

http://fedex-redelivery-confirm.com/pay

Your package will be returned to sender if no action is
taken within 48 hours.

FedEx Delivery Support

Analysis — Email 3 (Phishing):

Indicator	Detail	Risk
Sender Domain	`fedex-tracking-notification.com` — not the real `fedex.com`	🔴 High
Suspicious Link	`fedex-redelivery-confirm.com` — fake payment page	🔴 High
Payment Request	Asking for credit card details for a "$1.99 fee" — card harvesting	🔴 High
Urgency Language	"48 hours or returned to sender"	🟡 Medium
Generic Greeting	"Dear Customer" — no tracking number or name	🟡 Medium

Email Sample 4 — Legitimate Email (Comparison)

From: no-reply@github.com
To: developer@gmail.com
Subject: [GitHub] Please verify your email address

Hi developer,

Please verify your email address to finish setting up your GitHub account.

https://github.com/users/verify?token=abc123xyz

This link will expire in 24 hours.

If you did not create a GitHub account, you can safely ignore this email.

Thanks,
The GitHub Team
github.com

Analysis — Email 4 (Safe):

Indicator	Detail	Risk
Sender Domain	`github.com` — matches official domain ✅	✅ Safe
Link Domain	`github.com/users/verify` — matches official domain ✅	✅ Safe
Personalised Greeting	Addresses user by username ✅	✅ Safe
No Threats	No account suspension threats ✅	✅ Safe
Opt-Out Option	"If you didn't create an account, ignore this" ✅	✅ Safe

COMMON PHISHING INDICATORS

The following indicators were consistently observed across all phishing email samples analyzed in this report:

1. FAKE OR SPOOFED SENDER DOMAINS
   Phishing emails use domains that closely resemble legitimate ones.
   Example: bankofamerica-alert.com instead of bankofamerica.com
   Always check the full sender email address, not just the display name.

2. URGENCY AND FEAR-BASED LANGUAGE
   Phrases like "Your account will be locked", "Act within 24 hours", or "Immediate action required" are designed to bypass rational thinking and force quick, unthinking responses.

3. SUSPICIOUS OR MISMATCHED LINKS
   Hover over any link before clicking. If the URL shown in the browser does not match the company's official domain, it is likely a phishing link.
   Example: fedex-redelivery-confirm.com is not fedex.com

4. GENERIC GREETINGS
   Legitimate companies address you by your real name.
   "Dear Customer", "Dear User", or "Dear Valued Member" are strong indicators of mass phishing campaigns.

5. REQUESTS FOR SENSITIVE INFORMATION
   Legitimate organizations will NEVER ask for passwords, OTPs, credit card numbers, or personal details via email.

6. UNSOLICITED ATTACHMENTS
   Unexpected attachments — especially .zip, .exe, .doc, or .pdf files — may contain malware. Never open attachments from unknown senders.

7. POOR GRAMMAR AND SPELLING
   Many phishing emails contain grammatical errors, awkward phrasing, or unusual formatting that real companies would not send.

8. MISMATCHED BRANDING
   Logos that look slightly off, wrong colors, or outdated branding can indicate a spoofed email template.

RISK CLASSIFICATION SUMMARY

#	Email Sample	Classification	Risk Level
1	Fake Bank Alert	PHISHING	HIGH
2	Fake IT Support	PHISHING	HIGH
3	Fake Package Delivery	PHISHING	HIGH
4	GitHub Verification Email	SAFE	NONE

CLASSIFICATION CRITERIA

SAFE
- Sender domain matches official company domain
- Link URLs match official domain
- No urgency or threatening language
- Personalized greeting

SUSPICIOUS
- One or two minor red flags present
- Unusual sender domain but no malicious link
- Requires further investigation before acting

PHISHING
- Fake or spoofed sender domain
- Suspicious or malicious links
- Urgency, fear, or pressure tactics
- Requests for sensitive information

PREVENTION GUIDELINES FOR EMPLOYEES & BUSINESSES

FOR INDIVIDUAL USERS

1. Always verify the sender's email address
   Check the full domain — not just the display name.
   Hover over the sender name to reveal the real address.

2. Never click links directly from emails
   Instead, open a new browser tab and manually type
   the official website address.

3. Verify urgent requests through a separate channel
   If your "bank" emails you asking for verification,
   call the bank directly using the number on their
   official website — not the one in the email.

4. Enable Multi-Factor Authentication (MFA)
   Even if a password is stolen, MFA prevents attackers
   from accessing the account without a second verification step.

5. Keep software and browsers updated
   Updates patch known vulnerabilities that phishing
   links may try to exploit.

FOR BUSINESSES & ORGANISATIONS

1. Implement Email Filtering (SPF, DKIM, DMARC)
   These email authentication protocols prevent spoofed
   emails from reaching employee inboxes.

2. Run Regular Phishing Simulations
   Test employees with simulated phishing emails to
   measure awareness and identify training gaps.

3. Conduct Security Awareness Training
   Educate employees on how to identify phishing emails
   at least twice a year.

4. Create a Clear Reporting Process
   Employees should know exactly who to contact
   (e.g. IT Security team) when they receive a
   suspicious email — without fear of judgment.

5. Use a Password Manager
   Prevents employees from entering credentials on
   fake websites, as password managers only autofill
   on verified legitimate domains.

EMPLOYEE DO'S AND DON'TS

DO'S

✔ Verify the sender's full email address before responding
✔ Hover over links to check the real URL before clicking
✔ Report suspicious emails to your IT/security team immediately
✔ Use Multi-Factor Authentication on all accounts
✔ Access sensitive accounts by typing the URL directly in your browser
✔ Keep your devices and software up to date
✔ Use strong, unique passwords for every account
✔ Trust your instincts — if something feels wrong, it probably is

DON'TS

✘ Don't click links in unsolicited or unexpected emails
✘ Don't open attachments from unknown senders
✘ Don't share passwords, OTPs, or personal data via email
✘ Don't act on urgency — take a moment to verify first
✘ Don't use public Wi-Fi to access sensitive accounts
✘ Don't ignore security awareness training
✘ Don't assume an email is safe because it looks professional
✘ Don't reply to suspicious emails — even to unsubscribe

CONCLUSION

Phishing attacks are among the most effective and widely used tactics in a cybercriminal's toolkit. They succeed not by breaking through firewalls or exploiting software — but by exploiting human behavior.

This report has demonstrated that phishing emails share consistent, identifiable patterns — fake domains, urgency tactics, suspicious links, and generic greetings. Once employees learn to recognize these patterns, they become the organization's strongest line of defense.

The most effective protection against phishing is a combination of:

- Technical controls (email filtering, MFA, DNS protection)
- Human awareness (regular training and simulations)
- Clear reporting processes (so threats are caught quickly)

Security is not just an IT problem — it is everyone's responsibility.

DISCLAIMER

This report was prepared as part of the Future Interns
Cyber Security Internship — Task 2.

All phishing email samples used in this report are fictional
and were created solely for educational and awareness purposes.
They are based on real-world phishing patterns documented by
security researchers and public datasets.

No real email accounts, systems, or individuals were targeted
or harmed during this exercise.

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